GPU Speed Of Light Throughput

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor. High-level overview of the utilization for compute and memory resources of the GPU presented as a roofline chart.

Compute (SM) Throughput [%]	32.82	Duration [ms]	3.57
Memory Throughput [%]	19.40	Elapsed Cycles [cycle]	4972286
L1/TEX Cache Throughput [%]	24.89	SM Active Cycles [cycle]	3887242.27
L2 Cache Throughput [%]	14.33	SM Frequency [Ghz]	1.39
DRAM Throughput [%]	0.79	DRAM Frequency [Ghz]	9.49

Latency Issue This workload exhibits low compute throughput and memory bandwidth utilization relative to the peak performance of this device. Achieved compute throughput and/or memory bandwidth below 60.0% of peak typically indicate latency issues. Look at ▸ Scheduler Statistics and ▸ Warp State Statistics for potential reasons.

▸ Key Performance Indicators

Compute Throughput Breakdown

SM: Pipe Tensor Cycles Active [%]	32.82
SM: Inst Executed Pipe Lsu [%]	11.96
SM: Issue Active [%]	11.80
SM: Inst Executed [%]	11.80
SM: Mio2rf Writeback Active [%]	9.49
SM: Pipe Alu Cycles Active [%]	5.80
SM: Pipe Fma Cycles Active [%]	4.56
SM: Inst Executed Pipe Xu [%]	4.36
SM: Mio Inst Issued [%]	4.07
SM: Pipe Fmaheavy Cycles Active [%]	3.10
SM: Mio Pq Read Cycles Active [%]	0.54
SM: Mio Pq Write Cycles Active [%]	0.54
SM: Inst Executed Pipe Uniform [%]	0.32
SM: Inst Executed Pipe Adu [%]	0.26
SM: Inst Executed Pipe Cbu Pred On Any [%]	0.00
SM: Inst Executed Pipe Tex [%]	0
SM: Instruction Throughput Internal Activity [%]	0
IDC: Request Cycles Active [%]	0
SM: Memory Throughput Internal Activity [%]	0
SM: Pipe Fp64 Cycles Active [%]	0
SM: Inst Executed Pipe Ipa [%]	0

Memory Throughput Breakdown

L1: Data Pipe Lsu Wavefronts [%]	19.40
L1: Lsu Writeback Active [%]	18.99
L2: T Sectors [%]	14.33
L2: Lts2xbar Cycles Active [%]	14.27
L1: Lsuin Requests [%]	11.96
L1: Data Bank Reads [%]	9.25
L1: M Xbar2l1tex Read Sectors [%]	8.24
L2: D Sectors [%]	3.67
GPU: Compute Memory Access Throughput Internal Activity [%]	3.62
L2: T Tag Requests [%]	3.58
L2: Xbar2lts Cycles Active [%]	1.84
L1: M L1tex2xbar Req Cycles Active [%]	1.06
L1: Data Bank Writes [%]	1.05
DRAM: Cycles Active [%]	0.79
DRAM: Dram Sectors [%]	0.68
L2: D Sectors Fill Device [%]	0.28
L1: Texin Sm2tex Req Cycles Active [%]	0.02
L2: D Sectors Fill Sysmem [%]	0.00
L1: Tex Writeback Active [%]	0
L2: D Atomic Input Cycles Active [%]	0
L1: F Wavefronts [%]	0
L1: Data Pipe Tex Wavefronts [%]	0
GPU: Compute Memory Request Throughput Internal Activity [%]	0

Floating Point Operations Roofline
Performance [FLOP/s] (1 = 1e+12)
Arithmetic Intensity [FLOP/byte]
Floating Point Operations Roofline (Double Precision)
Performance [FLOP/s] (1 = 1e+12)
Arithmetic Intensity [FLOP/byte]
Floating Point Operations Roofline (Half Precision)
Performance [FLOP/s] (1 = 1e+13)
Arithmetic Intensity [FLOP/byte]
Floating Point Operations Roofline (Single Precision)
Performance [FLOP/s] (1 = 1e+12)
Arithmetic Intensity [FLOP/byte]
Tensor Core Operations Roofline
Performance [OP/s] (1 = 1e+14)
Arithmetic Intensity [OP/byte]

	# Operations	# Operations / Cycle	# Operations / s	Peak %	Peak Operations / Cycle	Peak Operations / s
Src:bf16 Dst:fp32 Sparsity:	137438953472	27554.90	38547889462299.97	65.63	41984	58733455227564.41
Src:bf16 Dst:fp32 Sparsity:	0	0	0	0	83968	117466910455128.83
Src:fp16 Dst:fp16 Sparsity:	0	0	0	0	83968	117466910455128.83
Src:fp16 Dst:fp16 Sparsity:	0	0	0	0	167936	234933820910257.66
Src:fp16 Dst:fp32 Sparsity:	0	0	0	0	41984	58733455227564.41
Src:fp16 Dst:fp32 Sparsity:	0	0	0	0	83968	117466910455128.83
Src:fp64	0	0	0	0	289.54	405058311914.24
Src:int1	0	0	0	0	1343488	1879470567282061…
Src:int4 Sparsity:off	0	0	0	0	335872	469867641820515.31
Src:int4 Sparsity:on	0	0	0	0	671744	939735283641030.62
Src:int8 Sparsity:off	0	0	0	0	167936	234933820910257.66
Src:int8 Sparsity:on	0	0	0	0	335872	469867641820515.31
Src:tf32 Dst:fp32 Sparsity:c	0	0	0	0	20992	29366727613782.21
Src:tf32 Dst:fp32 Sparsity:c	0	0	0	0	41984	58733455227564.41

PM Sampling

Timeline view of PM metrics sampled periodically over the workload duration. Data is collected across multiple passes. Use this section to understand how workload behavior changes over its runtime.

Maximum Sampling Interval [us]	2	# Pass Groups	2
Maximum Buffer Size [Mbyte]	23.07	-	-

Compute Workload Analysis

Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

Executed Ipc Elapsed [inst/cycle]	0.47	SM Busy [%]	32.82
Executed Ipc Active [inst/cycle]	0.61	Issue Slots Busy [%]	11.80
Issued Ipc Active [inst/cycle]	0.61		

Pipe Utilization (% of elapsed cycles)
Tensor (All)
Tensor (FP)
ALU
FMA
FP64
Tensor (INT)
Utilization [%]
Pipe Utilization (% of peak instructions executed over elapsed cycles)
Tensor (FP)
LSU
ALU
FMA
XU
Uniform
ADU
CBU
FMA (FP16)
FP64 (DMMA)
FP64 (FP64)
TEX
Tensor (INT)
Utilization [%]
Pipe Utilization (% of active cycles)
Pipe Utilization (% of peak instructions executed over active cycles)

Memory Workload Analysis

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed chart of the memory units. Detailed tables with data for each memory unit.

Memory Throughput [Gbyte/s]	7.22	Mem Busy [%]	19.40
L1/TEX Hit Rate [%]	0	Max Bandwidth [%]	18.99
L2 Hit Rate [%]	98.07	Mem Pipes Busy [%]	11.96
L2 Compression Input Sectors [sector]	0	Local Memory Spilling Requests	0
L2 Compression Ratio	0	Local Memory Spilling Request Overhead [%]	0
L2 Compression Success Rate [%]	0	L2 Persisting Size [Mbyte]	1.18

Shared Memory

	Instructions	Requests	Wavefronts	% Peak	Bank Conflicts
Shared Load	0	0	75530240	18.47	0
Shared Load Matrix	18882560	18882560			
Shared Store	16384	16384	65536	0.00	0
Shared Store From Global Lc	2113536	2113536	8454144	0.52	0
Shared Atomic	0	0	0	0	0
Other	-	-	3025193	0.75	1628
Total	21012480	21012480	87075113	19.74	1628

L1/TEX Cache

	Instructions	Requests	Wavefronts	% Peak	Sectors	Sectors/Req	Hit Rate	Bytes	Sector
Local Load	0	0	0	0	0	0	0	0	
Global Load	8192	8192	2106294	0.51	131072	16	0	4194304	
Global Load To Shared Store (acce	0	0			0	0		0	
Global Load To Shared Store (bypa	2113536	2097152			33554432	16	-	1073741824	
Surface Load	0	0	0	0	0	0	0	0	
Texture Load	0	0	0	0	0	0	0	0	
Global Store	8192	8192	8648	0.00	131072	16	0	4194304	
Local Store	0	0	0	0	0	0	0	0	
Surface Store	0	0	0	0	0	0	0	0	
Global Reduction	0	0	0	0	0	0	0	0	
Surface Reduction	0	0	0	0	0	0	0	0	
Global Atomic ALU	0	0	0	0	0	0	0	0	
Global Atomic CAS	0								
Surface Atomic ALU	0	0	0	0	0	0	0	0	
Surface Atomic CAS	0								
Loads	2121728	2105344	2106294	0.51	33685504	16	0	1077936128	
Stores	8192	8192	8648	0.00	131072	16	0	4194304	
Atomics & Reductions	0	0	0	0	0	0	0	0	

L2 Cache

	ors	Sectors/Req	% Peak	Hit Rate	Bytes	Throughput	ector Misses to Devic	ector Misses to Systen	Sector Misses to Peer
L1/TEX Load	33685504	4	14.26	98.05	1077936128	302331774652.44	655360	0	0
L1/TEX Store	131072	4	0.06	100	4194304	1176388228.22	0	0	0
L1/TEX Atomic ALU	0	0	0	0	0	0	0	0	0
L1/TEX Atomic CAS	0	0	0	0	0	0	0	0	0
L1/TEX Reduction	0	0	0	0	0	0	0	0	0
L1/TEX Total	33816576	4	14.32	98.06	1082130432	303508162880.66	655360	0	0
GCC Total	5376	4	0.00	85.94	172032	48250298.42	756	0	0
ECC Total	0	-	0	-	0	0	0	-	-

L2 Cache Eviction Policies

	First	Hit Rate	Last	Hit Rate	Normal	Hit Rate	Normal Demote	Hit Rate
L1/TEX Load	0	0	0	0	33685504	98.05	0	0
L1/TEX Store	0	0	0	0	131072	100	0	0
L1/TEX Atomic	0	0	0	0	0	-	0	-
L1/TEX Total	0	0	0	0	33816576	98.06	0	0
GPU Total	12512	53.81	0	0	33825422	98.06	0	0

Device Memory

	Sectors	% Peak	Bytes	Throughput
Load	656144	0.65	20996608	5888977642.95
Store	148536	0.15	4753152	1333129897.06
Total	804680	0.79	25749760	7222107540.01

Scheduler Statistics

Summary of the activity of the schedulers issuing instructions. Each scheduler maintains a pool of warps that it can issue instructions for. The upper bound of warps in the pool (Theoretical Warps) is limited by the launch configuration. On every cycle each scheduler checks the state of the allocated warps in the pool (Active Warps). Active warps that are not stalled (Eligible Warps) are ready to issue their next instruction. From the set of eligible warps the scheduler selects a single warp from which to issue one or more instructions (Issued Warp). On cycles with no eligible warps, the issue slot is skipped and no instruction is issued. Having many skipped issue slots indicates poor latency hiding.

Active Warps Per Scheduler [warp]	2.00	No Eligible [%]	84.83
Eligible Warps Per Scheduler [warp]	0.21	One or More Eligible [%]	15.17
Issued Warp Per Scheduler	0.15		

Warps Per Scheduler
GPU Maximum Warps Per Scheduler
Theoretical Warps Per Scheduler
Active Warps Per Scheduler
Eligible Warps Per Scheduler
Issued Warp Per Scheduler
0.0
4.0
8.0
12.0
16.0

Warp State Statistics

Analysis of the states in which all warps spent cycles during the kernel execution. The warp states describe a warp's readiness or inability to issue its next instruction. The warp cycles per instruction define the latency between two consecutive instructions. The higher the value, the more warp parallelism is required to hide this latency. For each warp state, the chart shows the average number of cycles spent in that state per issued instruction. Stalls are not always impacting the overall performance nor are they completely avoidable. Only focus on stall reasons if the schedulers fail to issue every cycle. When executing a kernel with mixed library and user code, these metrics show the combined values.

Warp Cycles Per Issued Instruction [cycle]	13.21	Avg. Active Threads Per Warp	32
Warp Cycles Per Executed Instruction [cycle]	13.21	Avg. Not Predicated Off Threads Per Warp	28.85

Instruction Statistics

Statistics of the executed low-level assembly instructions (SASS). The instruction mix provides insight into the types and frequency of the executed instructions. A narrow mix of instruction types implies a dependency on few instruction pipelines, while others remain unused. Using multiple pipelines allows hiding latencies and enables parallel execution. Note that 'Instructions/Opcode' and 'Executed Instructions' are measured differently and can diverge if cycles are spent in system calls.

Executed Instructions [inst]	193054720	Avg. Executed Instructions Per Scheduler [inst]	588581.46
Issued Instructions [inst]	193064928	Avg. Issued Instructions Per Scheduler [inst]	588612.59

NVLink Topology

NVLink Topology diagram shows logical NVLink connections with transmit/receive throughput.

NVLink Topology

The system does not have any NVLink connections.

NVLink Tables

Detailed tables with properties for each NVLink.

Logical NVLink Properties

The system does not have any NVLink connections.

NUMA Affinity

Non-uniform memory access (NUMA) affinities based on compute and memory distances for all GPUs.

NUMA ID Table

GPU ID	GPU Name	NUMA ID by CPU Affini	CPU Affinity	NUMA ID by Memory Affinity
0	NVIDIA GeForce RTX 3090	0	0-29	0

Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	128	Function Cache Configuration	CachePreferShared
Registers Per Thread [register/thread]	199	Static Shared Memory Per Block [byte/block]	0
Block Size	256	Dynamic Shared Memory Per Block [Kbyte/block]	98.30
Threads [thread]	32768	Driver Shared Memory Per Block [Kbyte/block]	1.02
Waves Per SM	1.56	Shared Memory Configuration Size [Kbyte]	102.40
Uses Green Context	0	Stack Size	1024
# SMs [SM]	82	# TPCs	41
Enabled TPC IDs	all	-	-

Small Grid If you execute __syncthreads() to synchronize the threads of a block, it is recommended to have at least two blocks per multiprocessor (compared to the currently executed 1.6 blocks) This way, blocks that aren't waiting for __syncthreads() can keep the hardware busy.

▸ Key Performance Indicators

Tail Effect
Est. Speedup: 50.00% A wave of thread blocks is defined as the maximum number of blocks that can be executed in parallel on the target GPU. The number of blocks in a wave depends on the number of multiprocessors and the theoretical occupancy of the kernel. This kernel launch results in 1 full waves and a partial wave of 46 thread blocks. Under the assumption of a uniform execution duration of all thread blocks, this partial wave may account for up to 50.0% of the total runtime of this kernel. Try launching a grid with no partial wave. The overall impact of this tail effect also lessens with the number of full waves executed for a grid. See the Hardware Model description for more details on launch configurations.

▸ Key Performance Indicators

Occupancy

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	16.67	Block Limit Registers [block]	1
Theoretical Active Warps per SM [warp]	8	Block Limit Shared Mem [block]	1
Achieved Occupancy [%]	16.59	Block Limit Warps [block]	6
Achieved Active Warps Per SM [warp]	7.96	Block Limit SM [block]	16

Theoretical Occupancy
Est. Local Speedup: 83.33% The 2.00 theoretical warps per scheduler this kernel can issue according to its occupancy are below the hardware maximum of 12. This kernel's theoretical occupancy (16.7%) is limited by the number of required registers, and the required amount of shared memory.

▸ Key Performance Indicators

Impact of Varying Register Count Per Thread
% Occupancy
Registers Per Thread
Impact of Varying Block Size
% Occupancy
Block Size
Impact of Varying Shared Memory Usage Per Block
% Occupancy
Shared Memory Per Block
Impact of Varying Register Count Per Thread
# Warps
Registers Per Thread
Impact of Varying Block Size
# Warps
Block Size
Impact of Varying Shared Memory Usage Per Block
# Warps
Shared Memory Per Block

GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

Average SM Active Cycles [cycle]	3887242.27	Average L1 Active Cycles [cycle]	3887242.27
Average L2 Active Cycles [cycle]	1820639.62	Average SMSP Active Cycles [cycle]	3880218.66
Average DRAM Active Cycles [cycle]	268226.67	Total SM Elapsed Cycles [cycle]	409001428
Total L1 Elapsed Cycles [cycle]	409001428	Total L2 Elapsed Cycles [cycle]	236167824
Total SMSP Elapsed Cycles [cycle]	1636005712	Total DRAM Elapsed Cycles [cycle]	406129664

SMs Workload Imbalance
Est. Speedup: 17.29% One or more SMs have a much lower number of active cycles than the average number of active cycles. Maximum instance value is 22.18% above the average, while the minimum instance value is 35.87% below the average.

▸ Key Performance Indicators

SMSPs Workload Imbalance
Est. Speedup: 17.27% One or more SMSPs have a much lower number of active cycles than the average number of active cycles. Maximum instance value is 22.20% above the average, while the minimum instance value is 35.79% below the average.

▸ Key Performance Indicators

L1 Slices Workload Imbalance
Est. Speedup: 17.29% One or more L1 Slices have a much lower number of active cycles than the average number of active cycles. Maximum instance value is 22.18% above the average, while the minimum instance value is 35.87% below the average.

▸ Key Performance Indicators

Workload Distribution

	Average	Min	Max	Sum
SM Active Cycles	3887242.27	2492814	4995211	318753866
SMSP Active Cycle	3880218.66	2491673	4987484	1272711719
L1 Active Cycles	3887242.27	2492814	4995211	318753866
L2 Active Cycles	1820639.62	1799725	1842902	87390702
DRAM Active Cycle	268226.67	267520	269056	3218720

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

Branch Instructions [inst]	526336	Branch Efficiency [%]	100
Branch Instructions Ratio [%]	0.00	Avg. Divergent Branches [branches]	0

Warp Stall Sampling (All Samples)

Location	Value	Value (%)
triton_impl.py:112 (0x75f54…	5,008	3
triton_impl.py:124 (0x75f54…	4,275	3
triton_impl.py:124 (0x75f54…	4,172	3
triton_impl.py:124 (0x75f54…	3,664	2
triton_impl.py:124 (0x75f54…	3,606	2

Most Instructions Executed

Location	Value	Value (%)
triton_impl.py:102 (0x75f54…	262,144	0
triton_impl.py:189 (0x75f54…	262,144	0
triton_impl.py:189 (0x75f54…	262,144	0
triton_impl.py:189 (0x75f54…	262,144	0
triton_impl.py:189 (0x75f54…	262,144	0

To customize your report even further, you might want to learn about custom sections and writing your own rules.
You might also want to consider adding individual metrics.